

NLU (natural language understanding) is an active research area and growing very rapidly, I would suggest that they allocate a subset of their data science talent to acquire new skills in this area.

Q. What aspects should decision-makers be wary of when dealing with Gen AI?

A. They must be mindful of two crucial aspects — first, identifying the business success metric they intend to achieve through the implementation of Gen AI-based offerings; and second, determining the criteria for what ‘good’ is. For instance, in the development of a question-answering system that integrates multiple data sources, it is essential to establish metrics to evaluate its goodness. Defining what ‘good’ looks like and establishing clear evaluation criteria are pivotal in ensuring the success of such endeavours.

Q. Besides the popular SaaS platforms, which traditional business applications are being affected/benefited by Gen AI?

A. Gen AI is impacting various sectors, with significant influence observed in education and the social sector. New patterns are emerging on how to solve a problem, and conversational interfaces are becoming first class citizens. However, health and finance will have their challenges due to the heavy regulation vis-a-vis the probabilistic nature of this technology. Gen AI implementations will have human augmentation in most places with the necessary guard rails.

Another major area of adoption of Gen AI is in the software development life cycle (SDLC). Apart from that, any creative field will have a major impact, be it music, arts, or writing, which is bound to create an interesting discussion on issues such as copyrights. Notably, Thoughtworks is also using Gen AI tech as a brainstorming partner with the necessary guard rails, and most of our customers are looking to adopt it. Gen AI

has also led to an acknowledgement about the need for having an AI strategy for the organisation. We have multiple offerings across literacy and education, use case evaluation, productionisation, and scaling.

Q. How do you see the role of open source in enabling growth of Gen AI?

A. The contributions from Meta, Hugging Face, and EleutherAI have been remarkable. Open source plays a crucial role in ensuring the ethical and responsible use of generative AI. Thanks to open source contributions, there is a strong belief that this technology will have a significant outreach to people and industries that may not find it economically feasible to adopt services from OpenAI, among others.

Q. What role does open source play in setting the standard for responsibility and ethics in the context of information dissemination in the case of generative AI tools such as ChatGPT?

A. We are currently in a catch-up game regarding these issues. For example, after the advent of ChatGPT, European countries formed a team to ensure that generative AI is used responsibly within Europe. Along with different types of licensing, more regulation is required by authoritative bodies. In India, we are yet to see significant efforts to regulate generative AI in various Indian languages; the government will need to play a role in framing regulations and laws around this technology. It's not just about leaving it to the people who built the technology; there needs to be a governing body. Many countries are forming committees for this purpose. In the US, there's a push for regulations that don't stifle competition. Open source communities are crucial here, as many enterprises are moving towards open source models, preferring to host their data and models independently rather than sending it to third parties.

The gap in regulation is acknowledged, but it will take time to bridge. Consumers of this technology now have multiple options and can make conscious decisions about which direction to go. Regulatory bodies, both independent and government-formed, are emerging to oversee and regulate these developments in the AI and open source sectors.

Q. Considering the potential unreliability of generative AI, as illustrated by an incident where ChatGPT created a fictitious quote attributed to a non-existent ‘John Doe’, how do we ensure the authenticity and reliability of the information generated by such AI systems?

A. Addressing the reliability of generative AI involves multiple levels and requires a degree of education about the correct use of the technology. The core issue is ensuring that responses are factual. This technology, however, wasn't designed to inherently distinguish fact from fiction; it's about predicting the next sequence of words based on its training. This is known as ‘closed book’ question-answering, where the AI responds based solely on its training data, without any external verification.

To improve reliability, one approach is ‘open book’ question-answering. This involves grounding the AI's responses in factual data. For instance, in an organisational context, you could feed the AI relevant PDFs or documents. When the AI generates a response, it's restricted to use only the information from these provided documents, ignoring its general training data. This method, known as retrieval-augmented generation (RAG), combines generative capabilities with data retrieval from specific, trusted sources.

However, it's important to recognise that generative AI is probabilistic, not deterministic. While these methods can reduce the incidence of incorrect or fictional information, they can't guarantee 100% accuracy. The goal